

Upload Dataset

Upload Loan Approval CSV file



loan_ap...ataset.csv
375.3KB



Loan Approval Prediction

Decision Tree Classifier

This application predicts whether a loan application will be Approved or Rejected.



Dataset

Shape of dataset: (4269, 13)

	loan_id	no_of_dependents	education	self_employed	income_annum	loan_amount	loan_term	cib
0	1	2	Graduate	No	9600000	29900000	12	
1	2	0	Not Graduate	Yes	4100000	12200000	8	
2	3	3	Graduate	No	9100000	29700000	20	
3	4	3	Graduate	No	8200000	30700000	8	
4	5	5	Not Graduate	Yes	9800000	24200000	20	
5	6	0	Graduate	Yes	4800000	13500000	10	
6	7	5	Graduate	No	8700000	33000000	4	
7	8	2	Graduate	Yes	5700000	15000000	20	
8	9	0	Graduate	Yes	800000	2200000	20	
9	10	5	Not Graduate	No	1100000	4300000	10	



Data Information

	Column	Missing
0	no_of_dependents	
1	education	
2	self_employed	
3	income_annum	
4	loan_amount	
5	loan_term	
6	cibil_score	
7	residential_assets_value	
8	commercial_assets_value	
9	luxury_assets_value	

Model Performance

Accuracy

97.89%

Precision

97.95%

Recall

97.89%

F1 Score

97.90%



Loan Application

Number of Dependents

2-+

CIBIL Score

750-+

Education

Graduatev

Residential Assets Value

5000000.00-+

Self Employed

Nov

Commercial Assets Value

2000000.00-+

Annual Income

8000000.00-+

Luxury Assets Value

3000000.00-+

Loan Amount

20000000.00-+

Bank Asset Value

4000000.00-+

Loan Term (Years)

15-+

Predict Loan Status

 Prediction Result

 LOAN APPROVED

 Prediction Probability

	Loan Status	Probability
0	Approved	100.00%
1	Rejected	0.00%

 Confusion Matrix

	Approved	Rejected
Approved	516	
Rejected	3	

 Decision Tree Information

Model: Decision Tree Classifier

Maximum Depth: 5

Minimum Samples Split: 10

Minimum Samples Leaf: 5

Test Size: 20%

Project Conclusion

The Decision Tree model predicts loan approval based on applicant financial and personal information.

The model can help perform an initial screening of loan applications.

Important factors include CIBIL score, income, loan amount and asset values.

The Streamlit application provides an easy interface for entering applicant details and getting a prediction.